

DocuBrain RAG Evaluation Report & Action Plan

Fullstack RAG Benchmark Analysis • NovaCloud Evaluation Dataset (20 Questions)

Total Questions: 20	Passed: 15 (75.0%)	Failed: 5 (25.0%)	LLM Engine: Gemini 3.5/3.8 Flash
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1. Completed Fix: Model Endpoint Upgrade (Action Plan 1)

Status: COMPLETED

We successfully updated all Gemini API model candidate lists across llm_service.py, rag_agent.py, tools.py, and agentic_chunker.py from deprecated models (gemini-2.5-flash / gemini-2.0-flash returning 404 errors) to active production models: **gemini-3.8-flash** and **gemini-3.5-flash**.

Result: The RAG system now successfully connects to live Gemini APIs, enabling full generative synthesis and autonomous tool calling without 404 endpoint failures.

2. Detailed Analysis of 5 Failing Questions

The table below details why 5 out of 20 benchmark questions failed and the underlying root causes:

ID & Category	Question & Expected Answer	Why it Failed (Root Cause)	Action Required
Q8 Conditional Reasoning	Q: If an application needs data to remain available after a regional failure, which NovaCloud capabilities help? Expected: Cross-region replication (NovaObject/NovaDocument), availability zones, NovaSQL read replicas.	Retrieval searched for 'regional failure' and pulled Page 1 (Regional Architecture) instead of Page 9 (Disaster Recovery & Cross-Region Replication).	Query Expansion / HyDE: Expand conditional queries to include terms like 'disaster recovery', 'regional outage', and 'cross-region replication'.
Q9 Timeline Retrieval	Q: When were Europe Central and Asia Pacific introduced? Expected: Europe Central in 2021, Asia Pacific in 2023.	Fixed character chunking (600 chars) sliced the answer sentence in half across Chunk #4 boundary (cut at '...Europe Centra').	Sentence-Aware Chunking: Increase chunk size to 800–1000 chars and enforce sentence boundary splitting.
Q10 Scenario Retrieval	Q: Temporary session storage that should not be only persistent copy? Expected: NovaCache (Redis-compatible, in-memory).	Chunk #13 was truncated right before the warning: 'Customers should not use NovaCache as the only persistent storage mechanism...'	Agentic / Semantic Chunking: Group text into conceptual spans rather than fixed character limits.

Q15 Distractor Test	Q: Does NovaCache provide persistent storage for critical data? Expected: No, it is in-memory only.	Retrieval missed the negative assertion warning sentence because paragraph context was severed by standard chunking.	Overlap & Negative Prompting: Increase overlap to 150 chars and instruct LLM to explicitly assert 'No' on distractor queries.
Q18 Precision / Contradiction	Q: Does NovaCloud automatically guarantee RPO of 15 min for all apps? Expected: No, NovaCloud does not guarantee RPO/RTO for all services.	The section concluding sentence ('NovaCloud does not automatically guarantee...') was truncated at the end of Chunk #32.	Chunk Overlap Increase: Expand overlap from 80 to 150 chars to retain paragraph concluding sentences.

3. Strategic Roadmap to Achieve 100% Pass Rate

- **1. Implement Query Expansion / HyDE:** Intercept user queries with semantic synonyms before vector search to solve conditional reasoning misses (Q8).
- **2. Upgrade Chunking Strategy to Agentic / Semantic:** Replace fixed character splitting with semantic_chunker_service or agentic_chunker_service to prevent sentence fragmentation (Q9, Q10).
- **3. Expand Chunk Size & Overlap:** Increase CHUNK_SIZE from 600 to 900 characters and CHUNK_OVERLAP from 80 to 150 characters to capture concluding caveats (Q15, Q18).
- **4. Add API Quota Retry Backoff:** Add retry delays for batch evaluation scripts to prevent 429 RESOURCE_EXHAUSTED quota limits on free-tier Gemini API keys.